

ECON 0100 | Vignette B2 | Supply | Solutions

Solution guide for the teaching team. Answers and working are in red; everything in black is what students see. Q1 is the consumer surplus practice Vignette B1 did not reach.

Q1 | Consumer Surplus

Members of the wizarding world have preferences for pumpkin pasties according to the following demand curve:

$$P_d = 12 - \frac{1}{2}Q_d$$

Prices are in galleons and quantity is in pasties.

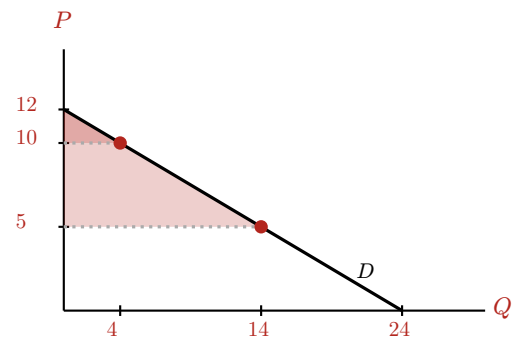
- a) Plot and find the area of consumer surplus at a price of 10 galleons. 4 galleons
- b) Plot and find the area of consumer surplus at a price of 5 galleons. 49 galleons
- c) How much did consumer surplus change as the price dropped from 10 to 5? up by 45 galleons

Consumer surplus is the triangle below demand and above the price, out to the quantity bought. First find the quantity: at 10 galleons, $10 = 12 - \frac{1}{2}Q_d$ gives $Q_d = 4$; at 5 galleons, $5 = 12 - \frac{1}{2}Q_d$ gives $Q_d = 14$. The triangle's height is the intercept minus the price and its base is the quantity.

a) $\frac{1}{2} \cdot 4 \cdot (12 - 10) = 4$ galleons.

b) $\frac{1}{2} \cdot 14 \cdot (12 - 5) = 49$ galleons.

c) $49 - 4 = 45$ galleons more. A lower price means every pasty already being bought keeps more surplus, and buyers add pasties whose marginal benefit was below 10 but above 5.



Q2 | Quantity Supplied

Pumpkin pasties are produced by many sellers according to the following supply curve:

$$P = 2 + \frac{1}{2}Q_s$$

Prices are in galleons and quantity is in pasties.

- a) Find and plot the quantity supplied at a price of 10 galleons. 16 pasties
- b) Find and plot the quantity supplied at a price of 5 galleons. 6 pasties
- c) How much did quantity supplied change as the price dropped from 10 to 5? down by 10 pasties
- d) What is marginal cost at 8 pasties? 6 galleons

The vertical intercept is 2 galleons: below that price nobody sells. Price to quantity: a) $10 = 2 + \frac{1}{2}Q_s$ gives $Q_s = 16$; b) $5 = 2 + \frac{1}{2}Q_s$ gives $Q_s = 6$; c) $6 - 16 = -10$, so quantity supplied fell by 10 pasties, which is the law of supply. Quantity to price: d) the height of the supply curve at 8 pasties is $2 + \frac{1}{2} \cdot 8 = 6$ galleons, the marginal cost of the eighth pasty.

Q3 | Producer Surplus

Use a graph to plot this supply curve, and find and label the producer surplus at both 5 galleons and 10 galleons.

PS at 5 galleons: 9 galleons

PS at 10 galleons: 64 galleons

- c) How much did producer surplus change as the price dropped from 10 to 5? down by 55 galleons

Producer surplus is the triangle above supply and below the price, out to the quantity sold, using the quantities from Q2. Its height is the price minus the 2-galleon intercept.

At 5 galleons: $\frac{1}{2} \cdot 6 \cdot (5 - 2) = 9$ galleons. At 10 galleons: $\frac{1}{2} \cdot 16 \cdot (10 - 2) = 64$ galleons.

c) $9 - 64 = -55$: producer surplus fell by 55 galleons. The revenue rectangle at 10 galleons is $10 \cdot 16 = 160$; the 64 above the curve is surplus and the 96 below it covers marginal costs.

